

Continuous-Space Concept Learning

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Mathematical Formalization

1. Perceptual Representation (The Object)

"What is X_i ?"

In this framework, a perceived object is not modeled as a deterministic point, but as a **probabilistic observation**[1,2]. This accounts for sensory noise and the inherent uncertainty of perception (e.g., a "fuzzy" location in color/shape space).

Let $\mathcal{Z} \subseteq \mathbb{R}^D$ be the feature space (where $D = 12$). For the i -th object in a scene, we observe a raw feature vector $\mathbf{x}_i$. We define the **Perceptual Distribution** O_i [1,2] as an isotropic Gaussian centered at $\mathbf{x}_i$:

$$P_{obj}(\mathbf{z} \mid O_i) = \mathcal{N}(\mathbf{z}; \mathbf{x}_i, \epsilon_{obj}\mathbf{I})$$

- $\mathbf{z} \in \mathbb{R}^D$: A latent variable in the feature space.
- $\mathbf{x}_i$: The observed mean vector (from `encoders.py`).
- ϵ_{obj} : A fixed, small variance hyperparameter (Code: `eps_obj`), representing the precision of visual grounding.

[1] **Ernst, M. O., & Banks, M. S. (2002)**. "Humans integrate visual and haptic information in a statistically optimal fashion." *Nature*.

(Abstract) Thus, the nervous system seems to combine visual and haptic information in a fashion that is similar to a maximum-likelihood integrator. Visual dominance occurs when the variance associated with visual estimation is lower than that associated with haptic estimation.

[2] **Knill, D. C., & Pouget, A. (2004)**. "The Bayesian brain: the role of uncertainty in neural coding and computation." *Trends in Neurosciences*.

(Page 1) The fundamental concept behind the Bayesian approach to perceptual computations is that the information provided by a set of sensory data about the world is represented by a conditional probability density function over the set of unknown variables – the posterior density function. A Bayesian perceptual system, therefore, would represent the perceived depth of an object, for example, not as a single number Z but as a conditional probability density function $p(Z/I)$, where I is the available image information (e.g. stereo disparities). Loosely speaking, $p(Z/I)$ would specify the relative probability that the object is at different depths Z , given the available sensory information.

2. Concept Representation (The Meaning)

"What is a concept? A set or a distribution?"

A concept C (e.g., the word "blue") is modeled as a **Continuous Probability Distribution**[3,4] over the feature space (Prototypical Theory), not a discrete set.

It is defined as a diagonal Gaussian distribution parameterized by a tuple $\theta_c = (\mu_c, \Sigma_c)$:

$$P_{conc}(\mathbf{z} \mid C) = \mathcal{N}(\mathbf{z}; \mu_c, \Sigma_c)$$

- μ_c (**Prototype**[3,4.**Quality Dimensions**]): The center of the concept (e.g., the ideal "blue").
- Σ_c (**Extension**[4.**Dimensional Attention**, 5]): A diagonal covariance matrix $\text{diag}(\sigma_1^2, \dots, \sigma_D^2)$.
 - *Small σ^2* : High specificity (relevant feature).
 - *Large σ^2* : Low relevance (the concept allows variation in this dimension).

[3] **Rosch, E. (1973)**. "Natural categories." *Cognitive Psychology*.

(Abstract) The hypothesis of the study was that the domains of color and form are structured into nonarbitrary, semantic categories which develop around perceptually salient 'natural prototypes.'

(P329 line 9) In contrast, many 'real' categories partition domains whose stimuli are not discrete but composed of continuous physical variations.

(P329 line 13) In most, if not all, natural language concepts, some stimuli are clearly better exemplars of the concept than others.

(P329 line 2) There are colors and forms which are more perceptually salient than other stimuli in their domains. A working hypothesis is that salient colors are those areas of the color space previously found to be most exemplary of basic color names in many different languages (Berlin & Kay, 1969) and that salient forms are the "good forms" of Gestalt psychology (circle, square, etc.). Such colors and forms more

[4] **Gärdenfors, P. (2000)**. *Conceptual spaces: The geometry of thought*. MIT Press.

(Section 1.3 Quality Dimensions) As first examples of quality dimensions, one can mention temperature, weight, brightness, pitch and the three ordinary spatial dimensions height, width, and depth. I have chosen these examples because they are closely* connected to what is produced by our sensory receptors (Schiffman 1982). The spatial dimensions height, width, and depth as well as brightness are perceived by the visual sensory system,⁵ pitch by the auditory system, temperature by thermal sensors and weight, finally, by the kinaesthetic sensors. As explained later in this chapter, however, there is also a wealth of quality dimensions that are of an abstract non-sensory character.

The primary function of the quality dimensions is to represent various "qualities" of objects.⁶ The dimensions correspond to the different ways stimuli are judged to be similar or different.⁷ In most cases, *judgments of similarity and difference generate an ordering relation* of stimuli. For example, one can judge tones by their pitch, which will generate an ordering from "low" to "high" of the perceptions.

The dimensions form the framework used to assign properties to objects and to specify relations among them. The coordinates of a point within a conceptual space represent particular instances of each dimension, for example, a particular temperature, a particular weight, and so forth. Chapter 3 will be devoted to how properties can be described with the aid of quality dimensions in conceptual spaces. The main idea is that a *property corresponds to a region* of a domain of a space.

(page 20 **Dimensional Attention**) "The book introduces the weighted Euclidean distance formulas (Equations 1.5 and 1.6), where w_i represents the attention weight assigned to dimension i ."

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The Euclidean and city-block metrics can be generalized in a straightforward way to the n -dimensional Cartesian space R^n by the following equations:

$$d_E(x, y) = \sqrt{\sum_i (x_i - y_i)^2}. \quad (1.2)$$

$$d_C(x, y) = \sum_i |x_i - y_i|. \quad (1.3)$$

They are special cases of the class of *Minkowski metrics* defined by

$$d_k(x, y) = \sqrt[k]{\sum_i |x_i - y_i|^k}, \quad (1.4)$$

where we thus have as special cases $d_E(x, y) = d_2(x, y)$ and $d_C(x, y) = d_1(x, y)$.²⁵

Equations (1.2)–(1.4) presume that the *scales* of the different dimensions are identical so that the distance measured along one of the axes is the same as that measured along another. In psychological contexts, however, this assumption is often violated. A more general form of distance is obtained by putting a weight w_i on the distance measured along dimension i (see, for example, Nosofsky 1986):

$$d_E(x, y) = \sqrt{\sum_i w_i \cdot (x_i - y_i)^2}. \quad (1.5)$$

$$d_C(x, y) = \sum_i w_i \cdot |x_i - y_i|. \quad (1.6)$$

In these equations, w_i is the “attention-weight” given to dimension i (the role of attention-weights in determining the “salience” of dimensions discussed in section 4.2). Large values of w_i “stretch” the conceptual space along dimension i , while small values of w_i will “shrink” the space along that dimension. In the following, I refer to the more general definitions given by equations (1.5) and (1.6) when Euclidean or city-block distances are mentioned.

[5] **Nosofsky, R. M. (1986).** "Attention, similarity, and the identification–categorization relationship." *Journal of Experimental Psychology: General*.

Nosofsky explicitly formulates "attention" as dimension weights within the distance metric, a concept analogous to the "weighted psychological space" found in models like INDSCAL:

$$d_{ij} = c \left(\sum_{k=1}^N w_k |x_{ik} - x_{jk}|^r \right)^{1/r}, \quad 0 < w_k < 1, \quad \sum_k w_k = 1$$

He provides a highly intuitive geometric interpretation for this set of parameters:

Geometric Interpretation: Weights cause the psychological space to "stretch or shrink" along the coordinate axes. A large weight (w_k) on a specific dimension implies that differences along that dimension are more "important," thereby expanding the distance between points in that direction.

Resource Constraint: The constraint ($\sum_k w_k = 1$) treats attention as a resource allocation problem: ignoring one dimension results in a "corresponding increase" in information extraction from another (at least within the model's parameterization).

Global Scale (c): This parameter represents overall discriminability (influenced by factors such as exposure or practice). It also relates to "whether the sharing of dimensional resources is fully compensatory" (the text discusses conditions where c falls below expectations, implying that the gain-loss trade-off may not always be perfectly compensatory).

3. Semantic Judgment (Inclusion Score)

"How do we judge if an object belongs to this concept?"

We define a continuous **truth value** (or compatibility score) based on the information-theoretic overlap between the object's distribution and the concept's distribution.

The **Log-Inclusion Score** is defined via the **Kullback-Leibler (KL) Divergence**:

$$\log_inc(O_i, C) = -\frac{1}{\tau} D_{KL} \left(P_{obj}(\cdot | O_i) \parallel P_{conc}(\cdot | C) \right)$$

Expanding the KL divergence for diagonal Gaussians:

$$\log_inc(O_i, C) = -\frac{1}{2\tau} \sum_{d=1}^D \left[\frac{(\mu_{c,d} - x_{i,d})^2}{\sigma_{c,d}^2} + \frac{\epsilon_{obj}}{\sigma_{c,d}^2} + \ln \left(\frac{\sigma_{c,d}^2}{\epsilon_{obj}} \right) - 1 \right]$$

- τ : Temperature parameter (Code: `tau`).
- **Significance**: If the object O_i is close to the prototype μ_c relative to the variance Σ_c , the penalty (KL) is low, and the score is high.

4. Pragmatic Inference (Rational Speech Acts)

We model communication as a recursive Bayesian inference process using the **Rational Speech Act (RSA)** framework (Frank & Goodman, 2012). To provide a unified theoretical basis for both single-referent and multi-referent communication, we adopt a **Utility-Based** perspective (Goodman & Frank, 2016).

This framework strictly delineates two computational roles:

1. **The Semantic Interpreter (L_0)**: Computes literal truth conditions and semantic precision (the "Size Principle") inherent to the language, independent of speaker effort.
2. **The Rational Speaker (S_1)**: Optimizes communicative utility by trading off **Informativeness** (reduction of listener uncertainty) against **Cost** (production effort).

A. Single-Intent Inference

This canonical model applies when the referent is a single target object $t \in \mathcal{O}$ (the set of visible objects).

1. Semantic Potential (The Literal Listener L_0)

The Literal Listener evaluates the semantic compatibility between a target object t and an utterance u . We define the **Semantic Potential**, $\phi(t, u)$, which integrates logical truth value and semantic precision. Crucially, this potential *excludes* utterance cost, as cost is a production constraint, not a comprehension constraint.

$$\ln \phi(t, u) = \underbrace{\sum_{w \in u} \log_inc(t, w)}_{\text{Truth Compatibility}} - \underbrace{\beta \sum_{w \in u} \text{Vol}(C_w)}_{\text{Semantic Precision}}$$

- $\log_inc(t, w)$: The log-inclusion score (inverted KL divergence) between object t and concept C_w .
- $\text{Vol}(C_w)$: The log-volume of the concept's variance. This term penalizes vague concepts (e.g., "thing") in favor of specific ones (e.g., "red"), implementing the **Size Principle**.
- β : Precision weighting hyperparameter.

The L_0 listener resolves contextual ambiguity by **normalizing** this potential over the domain of visible objects $\mathcal{O}$:

$$P_{L_0}(t | u) = \frac{\exp(\ln \phi(t, u))}{\sum_{t' \in \mathcal{O}} \exp(\ln \phi(t', u))}$$

2. The Rational Speaker (S_1)

The speaker is modeled as a rational agent selecting an utterance u from a set of alternatives $\mathcal{A}$ to maximize a **Utility Function** U_{S_1} . This function represents the trade-off between Informativeness and Cost.

$$U_{S_1}(u; t) = \alpha \cdot \underbrace{\ln P_{L_0}(t | u)}_{\text{Informativeness}} - \underbrace{\lambda \cdot |u|}_{\text{Effort Cost}}$$

- α : Rationality parameter (inverse temperature), controlling the optimality of the choice.
- λ : Cost per token, enforcing Grice's Maxim of Manner (brevity).

The speaker's policy is defined by a Softmax distribution over utilities:

$$P_{S_1}(u | t) = \frac{\exp(U_{S_1}(u; t))}{\sum_{u' \in \mathcal{A}} \exp(U_{S_1}(u'; t))}$$

3. The Pragmatic Listener (L_1)

The final inference inverts the speaker model using Bayes' rule, assuming a uniform prior $P(t)$ over objects:

$$P_{L_1}(t | u) \propto P_{S_1}(u | t) \cdot P(t)$$

B. Multi-Intent Inference (Cardinality & Sets)

For utterances specifying cardinality (e.g., "2 blue", "1 red, 1 solid"), the referential target generalizes to an **Assignment Set** $T \subseteq \mathcal{O}$. The goal is to identify the subset of objects T that satisfies the utterance's intent components $u = \{(k_j, W_j)\}_{j=1}^m$, where k_j is the requested count and W_j is the set of tokens (e.g., {blue, box}).

The system supports two distinct modes for defining the set-level semantic potential $\phi_{set}(T, u)$, serving different computational and theoretical goals.

Mode I: Differentiable Cardinality Relaxation ("Soft Count")

Assumption: Cardinality is a continuous semantic constraint suitable for gradient-based grounding.

In this regime, "exactly k " is treated as a soft constraint enforced by an auxiliary loss. This preserves differentiability, allowing the model to learn concept parameters end-to-end without discrete counting operations.

1. **Visual Presence:** We estimate the continuous "count" of a feature w within a set T using a sigmoid-gated sum:

$$\hat{N}(w, T) = \sum_{o \in T} \sigma(s \cdot (\log_inc(o, w) + \delta))$$

where s, δ are scaling and bias parameters centering the logical transition (e.g., $s = 0.2, \delta = 38.0$).

2. **Penalized Potential:** The semantic potential includes an **Exactness Penalty** $\mathcal{L}_{exact}$ (squared error) to enforce cardinality on every token:

$$\ln \phi_{soft}(T, u) = \left(\sum_{o \in T} \ln \phi(o, u) \right) - \gamma \sum_{j=1}^m \sum_{w \in W_j} (\hat{N}(w, T) - k_j)^2$$

This forces the semantic score to peak only when the aggregate visual evidence for every descriptive token aligns with the requested cardinality.

Mode II: Pure Pragmatic Competition ("Scalar Implicature")

Assumption: Cardinality arises from the pragmatic competition of numerals.

In this theoretically rigorous regime, numerals are treated as lower bounds ("at least k "), and exactness emerges purely from the S_1 competition (Scalar Implicature).

1. **Lower-Bound Semantics:** The semantic potential acts as a hard gate on set size, with a weak prior η to prevent trivial bias toward maximal sets:

$$\ln \phi_{pure}(T, u) = \left\{ \sum_{o \in T} \ln \phi(o, u) - \eta |T| \quad \text{if } \forall j, |T \cap \text{scope}(W_j)| \geq k_j \quad -\infty \quad \text{otherwise} \right.$$

2. **Implicature via Alternatives:** The set of alternatives $\mathcal{A}$ is augmented to include varying numerals (e.g., $\mathcal{A} \supseteq \{ "1 \text{ blue}", "2 \text{ blue}", "3 \text{ blue}" \}$).
 - **Logic:** If the target set size is $|T| = 3$, the utterance "3 blue" yields higher informativeness (P_{L_0}) than "2 blue" (which is semantically compatible with sizes 2, 3, and 4).
 - **Emergence:** The rational speaker S_1 prefers "3 blue". Consequently, upon hearing "2 blue", the listener L_1 infers that the target is **not** size 3.

3. **Probabilistic Counting:** Unlike Mode I, counting here is derived by marginalizing the posterior:

$$\mathbb{E}[|T|] = \sum_{o \in \mathcal{O}} \sum_{T: o \in T} P_{L_1}(T \mid u)$$

Unified Multi-Intent Speaker (S_1)

Regardless of the mode used for L_0 , the Speaker S_1 computes utility by normalizing over the space of valid Assignments $\mathcal{T}$ (e.g., all subsets of $\mathcal{O}$):

1. **Set-Level Normalization:**

$$P_{L_0}(T \mid u) = \frac{\exp(\ln \phi_{set}(T, u))}{\sum_{T' \in \mathcal{T}} \exp(\ln \phi_{set}(T', u))}$$

2. **Speaker Utility:**

$$U_{S_1}(u; T) = \alpha \ln P_{L_0}(T \mid u) - \lambda |u|$$

This formulation ensures that the speaker optimizes for distinguishing the target set T from *other possible sets*, naturally unifying the logic for both single and multiple referents.

5. Incremental Learning (The Update Rule)

"How does the concept update mathematically?"

Learning is modeled as **Online Bayesian Moment Matching (or Chan's Generalization of Welford's Algorithm[6])**. We treat the pragmatic inference results as "soft labels" to update the concept parameters.

Given a scene with objects $\{O_1, \dots, O_N\}$ and an utterance u :

1. **E-Step (Soft Weights):** Compute the probability that each object is the target.

$$w_i = k \cdot P_{L_1}(t = O_i \mid u)$$

(Where k is the cardinality, e.g., 1).

2. **M-Step (Parameter Update):** Update the concept (μ, Σ, κ) using weighted batch statistics.

Let $W = \sum w_i$ be the total weight, and $\bar{\mathbf{x}}_{batch} = \frac{1}{W} \sum w_i \mathbf{x}_i$ be the batch mean.

- **Update Count:**

$$\kappa' = \kappa + W$$

- **Update Prototype (μ):**

The mean moves towards the weighted average of the likely objects.

$$\mu' = \mu + \frac{W}{\kappa'} (\bar{\mathbf{x}}_{batch} - \mu)$$

- **Update Variance (Σ):**

The variance expands to cover the spread of the new data ($M_{2,batch}$) and the shift in the mean.

$$M'_2 = M_2 + \sum w_i (\mathbf{x}_i - \bar{\mathbf{x}}_{batch})^2 + \frac{\kappa W}{\kappa'} (\bar{\mathbf{x}}_{batch} - \mu)^2$$

$$\Sigma' = \frac{M'_2}{\kappa'}$$

[6] **Bishop, C. M. (2006).** *Pattern Recognition and Machine Learning*.

(See Chapter 9.2, Page438: Mixture Models and EM). Bishop provides a detailed derivation of the formulas for updating μ and Σ in Gaussian Mixture Models, utilizing "responsibilities" (weights) to weight the parameter updates.

6. Zero-Shot Learning from word embedding: Semantic-Visual Isomorphism

"How is a concept defined without visual training samples?"

Zero-Shot Learning is based on the **Structural Isomorphism Assumption**: The geometric relationships in the semantic embedding space (Language) are locally preserved in the continuous visual feature space (Perception).

If the semantic vector for "orange" lies between "red" and "yellow", the visual distribution for "orange" should be predictably synthesized from the visual distributions of "red" and "yellow". Instead of relying on heuristic convex interpolation (e.g., Nadaraya-Watson kernel regression via Softmax), we formalize this cross-modal mapping as a rigorous **Gaussian Process (GP) Regression**. This provides a principled probabilistic mechanism to synthesize not just the visual prototype (mean), but to strictly disentangle the uncertainty (variance) into epistemic and aleatoric components.

A. Gaussian Process on the Semantic Manifold

Let $\mathbf{v}_{new} \in \mathbb{R}^E$ be the normalized semantic embedding of a new, unseen token (e.g., "orange"). Let $\mathcal{A}$ be the set of N Anchor Concepts—concepts that have both learned visual parameters ($\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j$) and semantic embeddings $\mathbf{v}_j$.

We model the mapping from the semantic manifold $\mathcal{V}$ to the perceptual parameter space Θ using independent Gaussian Processes for each perceptual dimension $d \in \{1, \dots, D\}$, sharing a strictly positive-definite Radial Basis Function (RBF) kernel:

$$\mu_d(\mathbf{v}) \sim \mathcal{GP}(0, k(\mathbf{v}, \mathbf{v}'))$$

$$k(\mathbf{v}, \mathbf{v}') = \sigma_f^2 \exp\left(-\frac{|\mathbf{v} - \mathbf{v}'|^2}{2l^2}\right)$$

- σ_f^2 : **Signal variance** (Code: `signal_var`), representing the amplitude of the kernel (the prior variance).
- l : **Length scale** (Code: `length_scale`), dictating how rapidly the semantic influence decays over distance.

B. Parameter Synthesis (Mean and Variance Decomposition)

Given the anchor embeddings matrix $\mathbf{V} \in \mathbb{R}^{N \times E}$ and the target visual means $\mathbf{Y} \in \mathbb{R}^{N \times D}$, we evaluate the cross-kernel vector $\mathbf{k}_* = k(\mathbf{V}, \mathbf{v}_{new}) \in \mathbb{R}^N$ and the $N \times N$ training covariance matrix $\mathbf{K} = k(\mathbf{V}, \mathbf{V})$.

The GP projection weights $\boldsymbol{\alpha}$ are computed as:

$$\boldsymbol{\alpha} = (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*$$

- σ_n^2 : **Observation noise** (Code: `noise_var`), acting as a Tikhonov regularizer for numerical stability.

1. Synthesized Mean (The Prototype)

The visual prototype of the hallucinated concept is the GP posterior predictive mean:

$$\boldsymbol{\mu}_{new} = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{Y} = \boldsymbol{\alpha}^\top \mathbf{Y}$$

Unlike Softmax attention (which is strictly a convex combination where weights sum to 1), the GP weights $\boldsymbol{\alpha}$ can legitimately contain negative values. This mathematically allows the model to perform true geometric extrapolation rather than strictly interpolating within the convex hull of anchors.

2. Synthesized Variance (Uncertainty Decomposition)

A hallucinated concept must rigorously model its own unreliability. The total diagonal variance

$\Sigma_{new} = \text{diag}(\sigma_{new,1}^2, \dots, \sigma_{new,D}^2)$ is explicitly decomposed into three components:

$$\sigma_{new,d}^2 = \underbrace{s_{epi}^2(\mathbf{v}_{new})}_{\text{Epistemic}} + \underbrace{\sigma_{ale,d}^2(\mathbf{v}_{new})}_{\text{Aleatoric}} + \underbrace{\epsilon_{\min}}_{\text{Floor}}$$

- **Epistemic Uncertainty (s_{epi}^2):** Measures the model's structural ignorance due to the lack of nearby anchors in the semantic space. It is a scalar derived from the GP predictive variance and is shared across all perceptual dimensions:

$$s_{epi}^2(\mathbf{v}_{new}) = \max \left(k(\mathbf{v}_{new}, \mathbf{v}_{new}) - \boldsymbol{\alpha}^\top \mathbf{k}_*, 0 \right) = \max \left(\sigma_f^2 - \boldsymbol{\alpha}^\top \mathbf{k}_*, 0 \right)$$

- **Aleatoric Uncertainty ($\sigma_{ale,d}^2$):** Measures the intrinsic visual vagueness inherited from the anchors. A composite concept should inherit the fuzziness of its constituents. It is computed via GP-weighted interpolation of the anchors' own learned variances $\sigma_{j,d}^2$. To ensure valid probability measures from potentially negative $\boldsymbol{\alpha}$ values, we normalize their absolute magnitudes:

$$w_{ale,j} = \frac{|\alpha_j|}{\sum_{i=1}^N |\alpha_i| + 10^{-12}}$$

$$\sigma_{ale,d}^2(\mathbf{v}_{new}) = \sum_{j=1}^N w_{ale,j} \cdot \sigma_{j,d}^2$$

- **Variance Floor ($\epsilon_{\min}$):** A strict lower bound (Code: `var_floor`, typically 0.01) to prevent density collapse and ensure numerical stability during downstream KL-divergence computations in RSA.

C. Theoretical Foundation: GP vs. Kernel Interpolation

This GP-based formulation supersedes previous Nadaraya-Watson (Softmax) approaches by providing strict probabilistic guarantees:

1. **Out-of-Distribution Awareness & True Extrapolation:** Softmax weights always sum to 1. If queried with a completely irrelevant semantic vector, a Softmax interpolator will confidently generate an average of the anchors with falsely low variance. The GP model, instead, correctly detects low kernel activation, drops $\boldsymbol{\alpha} \rightarrow \mathbf{0}$, and heavily spikes $s_{epi}^2 \rightarrow \sigma_f^2$, yielding a maximally uncertain, flat distribution.
2. **Disentangled Uncertainty vs. Heuristic Inflation:** The previous model required an artificial variance inflation scalar (γ) to represent hallucination uncertainty. The GP framework natively separates the uncertainty of "not knowing what the word means" (Epistemic s_{epi}^2) from "the intrinsic visual vagueness of the underlying concepts" (Aleatoric $\sigma_{ale,d}^2$). This analytically justifies the zero-shot concept's variance without relying on ad-hoc scalar multipliers.

Through this formulation, Zero-Shot hallucinated concepts seamlessly integrate into the RSA Pragmatic Inference engine (Section 4). Their dynamically inflated epistemic variance naturally lowers their L_0 log-inclusion score (due to the volume penalty and wider density spread), causing the **Rational Speaker (S_1)** to appropriately avoid using zero-shot words if a more grounded, high-precision anchor word is available.

7. Metacognition: Distinguishing Known vs. Unknown

"How does the model know if it knows an object?"

The model uses an **Epistemic Familiarity Score** (or "Explainedness") to estimate its own uncertainty. Instead of forcing a classification, it evaluates the global confidence of its current knowledge base regarding a specific observation.

A. Familiarity Score Function (E)

Let $\mathcal{K}$ be the set of currently learned concepts. For a perceived object O_x (represented by feature vector $\mathbf{x}$), the familiarity score $E(\mathbf{x})$ is the aggregated evidence from all grounded concepts.

$$E(\mathbf{x}) = \sum_{C \in \mathcal{K}} \mathbb{I}(\log_{\text{inc}}(\mathbf{x}, C) > \delta) \cdot \log_{10}(1 + \kappa_C)$$

Where:

- δ (**Match Threshold**): A minimum log-probability threshold (Code: `match_threshold`, e.g., -44.0). If the object is statistically too far from concept C , the indicator function $\mathbb{I}(\cdot)$ is 0.
- κ_C (**Concept Maturity**): The number of times concept C has been trained.
- $\log_{10}(1 + \kappa_C)$: This term ensures that **mature concepts** contribute significantly to the familiarity score, while provisional or noisy concepts (where $\kappa \approx 0$) contribute negligible confidence.

B. The Decision Rule (Novelty Detection)

We define a **Novelty Threshold** θ_{novel} (Code: `novelty_threshold`). The system classifies the cognitive state of the object as:

$$\text{State}(\mathbf{x}) = \begin{cases} \text{KNOWN} & \text{if } E(\mathbf{x}) \geq \theta_{\text{novel}} \\ \text{UNKNOWN} & \text{if } E(\mathbf{x}) < \theta_{\text{novel}} \end{cases}$$

- **Logic**: If the score is below θ_{novel} , the agent recognizes the object as novel, creates a temporary token (e.g., `concept_a1b2`), and triggers the "Ask" mechanism.
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8. Neuro-Symbolic World Model (The Physics Engine)

(`experiment_action.pdf` experiment 1)

"How does the agent learn and predict environmental dynamics?"

We formalize the environment's physical laws not as static symbolic rules, but as **Joint Gaussian Concepts** over a continuous state-transition manifold. This bridges the gap between subsymbolic perception (pixels) and symbolic reasoning (rules), allowing the agent to perform probabilistic inference about causality.

A. The Causal Manifold

Let $\mathcal{S} \subseteq \mathbb{R}^D$ be the perceptual state space (where $D = 27$, representing a flattened 3×3 RGB grid). We define a physical rule $\mathcal{R}$ as a concept residing in the joint space $\mathcal{Z}_{rule} \subseteq \mathbb{R}^{2D}$.

A single observation of a physical event is a vector $\mathbf{z}_{event}$ formed by concatenating the initial state (Context) and the temporal difference (Effect):

$$\mathbf{z}_{event} = [\mathbf{s}_t^\top, \Delta \mathbf{s}_t^\top]^\top \in \mathbb{R}^{54}$$

where $\mathbf{s}_t$ is the encoded state vector at time t , and $\Delta \mathbf{s}_t = \mathbf{s}_{t+1} - \mathbf{s}_t$.

B. Probabilistic Physical Laws

A physical law k (e.g., "gravity" or "levitate") is modeled as a diagonal Gaussian distribution $P(\cdot \mid \mathcal{R}_k)$ over this joint space:

$$P(\mathbf{z} \mid \mathcal{R}_k) = \mathcal{N} \left(\mathbf{z}; \begin{bmatrix} \boldsymbol{\mu}_{ctx}^{(k)} \\ \boldsymbol{\mu}_{eff}^{(k)} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{ctx}^{(k)} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{eff}^{(k)} \end{bmatrix} \right)$$

- **Context $\boldsymbol{\mu}_{ctx}$:** Defines under what conditions the rule applies (e.g., "red object above green object").
- **Effect $\boldsymbol{\mu}_{eff}$:** Defines what happens (e.g., "red object moves down").
- **Variance $\boldsymbol{\Sigma}$:** Encodes generalization. High variance in a specific dimension implies that feature is irrelevant to the rule's activation or effect.

C. Bayesian Predictive Inference

To predict the next state $\hat{\mathbf{s}}_{t+1}$ given a current state $\mathbf{s}_t$, the agent performs Bayesian model selection over the set of learned rules $\{\mathcal{R}_k\}$.

1. **Rule Activation (Likelihood):** We calculate the log-inclusion score of the current state against the rule's context parameters. To align with our perceptual framework, the deterministic state $\mathbf{s}_t$ is treated as an observation O_t with isotropic sensor noise $\mathcal{N}(\mathbf{s}_t, \epsilon_{obj} \mathbf{I})$, and the rule's context is treated as a sub-concept $C_{ctx}^{(k)} = (\boldsymbol{\mu}_{ctx}^{(k)}, \boldsymbol{\Sigma}_{ctx}^{(k)})$:

$$S_k = \log_inc(O_t, C_{ctx}^{(k)})$$

2. **Posterior Mixing Weights:** The contribution of each rule is determined by a Softmax function parameterized by a physical temperature τ_{phys} (Code: `temp`):

$$w_k = \frac{\exp(S_k / \tau_{phys})}{\sum_j \exp(S_j / \tau_{phys})}$$

3. **Effect Synthesis:** The predicted delta is a precision-weighted mixture of the experts' expected effects:

$$\Delta \hat{\mathbf{s}} = \sum_k w_k \boldsymbol{\mu}_{eff}^{(k)}$$

4. **State Update:**

$$\hat{\mathbf{s}}_{t+1} = \mathbf{s}_t + \Delta \hat{\mathbf{s}}$$

9. Pixel Motor System (Constraint-Based Action Synthesis & Motor Babbling)

"How are actions acquired and represented in a continuous signal space?"

Instead of relying on pre-programmed symbolic actions (e.g., `move_to(4)`), we model the motor cortex as a generative system of **Sensorimotor Primitives** learned directly in the raw pixel space. The system evolves through two distinct phases: a developmental phase of **Motor Babbling** (stochastic acquisition) and a mature phase of **Action Synthesis** (zero-shot composition).

A. Developmental Learning: Motor Babbling

"How does the agent learn to disentangle 'what' from 'where'?"

The agent acquires grounding for atomic concepts through an unsupervised process of stochastic exploration, formally modeled as Conditional Monte Carlo Estimation. We define a generative motor function $G : \mathcal{C} \times \mathcal{P} \rightarrow \mathbb{R}^D$ ($D = 36$) that produces a visual state vector given color parameter $\mathbf{c}$ and position parameter p .

- **Representation (RGBI, $9 \times 4 = 36D$):** The 3×3 grid is flattened into 9 cells. Each cell has 4 channels [R, G, B, I]. I is an intensity / structure channel computed as: $I = \text{mean}(R, G, B) = (R + G + B) / 3$ (Assumption) R,G,B are normalized to $[0,1]$, hence $I \in [0,1]$.

The learning objective is to estimate the marginal distribution of each feature by randomizing nuisance variables.

Phase I: Chromatic Grounding (Learning "Color")

To learn a color concept C_{rgb} (e.g., "cmd_red"), the agent fixes the target chromatic vector $\mathbf{c}_{target}$ while sampling spatial positions p_i from a uniform distribution $\mathcal{U}(\mathcal{P})$ over the grid:

$$\mathcal{D}_{color} = \left\{ \mathbf{x}_i \mid \mathbf{x}_i = G(\mathbf{c}_{target}, p_i), p_i \sim \mathcal{U}(\mathcal{P}) \right\}_{i=1}^N$$

Phase II: Spatial Grounding (Learning "Position")

To learn a spatial concept C_{pos} (e.g., "cmd_pos_4"), the agent fixes the target coordinate p_{target} while sampling colors from the available palette $\mathcal{U}(\mathcal{C})$:

$$\mathcal{D}_{pos} = \left\{ \mathbf{x}_j \mid \mathbf{x}_j = G(\mathbf{c}_j, p_{target}), \mathbf{c}_j \sim \mathcal{U}(\mathcal{C}) \right\}_{j=1}^N$$

The concept parameters are updated incrementally via **Vectorized Online Moment Matching** (Chan's formulation of Welford's Algorithm). Let κ be the observation count and $\mathbf{m}_2$ be the sum of squared differences vector. For a new sample vector $\mathbf{x}$, the element-wise updates (using the Hadamard product $\odot$) are:

- **Count:** $\kappa' = \kappa + 1$
- **Mean:** $\boldsymbol{\mu}' = \boldsymbol{\mu} + \frac{\mathbf{x} - \boldsymbol{\mu}}{\kappa'}$
- **Sum of Squared Differences:** $\mathbf{m}_2' = \mathbf{m}_2 + (\mathbf{x} - \boldsymbol{\mu}) \odot (\mathbf{x} - \boldsymbol{\mu}')$
- **Variance:** $\boldsymbol{\Sigma}' = \text{diag} \left(\frac{\mathbf{m}_2'}{\kappa'} \right) + \epsilon \mathbf{I}$

B. Representation: Orthogonal Disentanglement via Variance

Through this stochastic process, the diagonal variance parameters naturally encode **Semantic Orthogonality**. Since the 27D space represents RGB values across 9 spatial locations, the variance vector σ^2 acts as a learned **Relevance Gate**:

- **Color Primitives (e.g., "Red")**: Variance approaches the noise floor ϵ for Green and Blue channels across *all* spatial locations (enforcing "Must NOT be Green/Blue"). Variance remains high for the Red channel across all locations (enforcing "Red can appear anywhere").
- **Position Primitives (e.g., "Pos 4")**: Variance approaches ϵ for all RGB channels at *non-target* locations (enforcing "Must be empty elsewhere"). Variance remains high for all RGB channels at the target location (enforcing "Can be any color here").

This structure gives each concept "veto power" (high precision) only over the dimensions relevant to its semantic definition, while remaining completely agnostic to others.

C. Zero-Shot Action Synthesis (Product of Experts)

"How does the agent execute 'Red at Position 4' without training on it?"

To execute a composite command $u = \{C_{red}, C_{pos4}\}$ never seen during training, the system computes the intersection of the constraints imposed by each primitive. Mathematically, the posterior action distribution is the **Product of Gaussian Experts (PoE)**:

$$P(\mathbf{a} | u) \propto \prod_{k \in u} \mathcal{N}(\mathbf{a}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Precision-Weighted Averaging (The Synthesis Formula):

Since the product of Gaussians is a Gaussian, the synthesized mean action $\boldsymbol{\mu}_{synth}$ is derived analytically. Let $\boldsymbol{\Lambda}_k = \boldsymbol{\Sigma}_k^{-1}$ be the precision matrix (diagonal).

$$\boldsymbol{\mu}_{synth} = \left(\sum_{k \in u} \boldsymbol{\Lambda}_k \right)^{-1} \left(\sum_{k \in u} \boldsymbol{\Lambda}_k \boldsymbol{\mu}_k \right)$$

Mechanism of Constraint Intersection:

- For RGB dimensions: $\Lambda_{red} \gg \Lambda_{pos}$, so the weighted average is strictly dominated by $\boldsymbol{\mu}_{red}$.
- For spatial dimensions: $\Lambda_{pos} \gg \Lambda_{red}$, so the weighted average is strictly dominated by $\boldsymbol{\mu}_{pos}$.

The result is a coherent vector satisfying both properties simultaneously (Logical AND).

Signal Amplification:

The intersection of high-dimensional probability densities often results in attenuated magnitudes due to the "conflict" in high-variance dimensions. We apply a heuristic renormalization (Code: `amplify`) to restore signal intensity for the motor actuator:

$$\mathbf{a}^* = \frac{\boldsymbol{\mu}_{synth}}{\max_d(\mu_{synth,d})}$$

10. Planning Agent (Embodied Planning via Hierarchical Bayes)

(experiment_action.pdf experiment 2)

"How does the agent generalize dynamics across visual attributes?"

The Planning Agent extends the world model to achieve **Systematic Generalization** across sensory domains (e.g., learning the dynamic concept "grow" on red objects and seamlessly applying it to blue ones). Instead of heuristically discarding chromatic information via feature whitening, we formalize this capability using a rigorous **Hierarchical Bayesian Mixed-Effects Model**. This approach preserves continuous feature fidelity while enabling robust Few-Shot and Zero-Shot generalization.

A. Hierarchical Dynamic Concepts (The Mixed-Effects Model)

To learn abstract dynamic concepts independent of specific object features, we treat each color context $c \in \mathcal{C}$ as a distinct task that shares a global structural prior.

Given a state change observation $\Delta \mathbf{s} \in \mathbb{R}^D$, the expected effect for a specific color context c is decomposed into a shared structural prototype and a color-specific residual:

$$\boldsymbol{\mu}_{\text{eff}, c} = \boldsymbol{\mu}_{\text{shared}} + \boldsymbol{\delta}_c$$

- **Shared Prototype ($\boldsymbol{\mu}_{\text{shared}}$):** The *Fixed Effect*. It captures the color-invariant topological transformation (e.g., "where the pixels expand"), accumulating statistical evidence across all training samples regardless of their colors.
- **Color-Specific Residual ($\boldsymbol{\delta}_c$):** The *Random Effect*. It captures the fine-grained chromatic nuances specific to color c .

To prevent overfitting and enable rapid generalization to novel colors, we impose a zero-mean isotropic Gaussian prior on these residuals:

$$\boldsymbol{\delta}_{*c} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

where the hyperparameter σ_δ^2 (Code: `sigma_delta`) dictates the strength of **Shrinkage**. A small σ_δ^2 heavily regularizes the residual towards zero, compelling the model to attribute most of the observed variance to the shared structural prior unless strong color-specific evidence is observed.

B. Few-Shot Adaptation and Zero-Shot Fallback

This hierarchical formulation provides a mathematically principled mechanism for generalization:

- **Zero-Shot Fallback:** If an entirely unseen color c' is encountered, the MAP estimate of the residual defaults exactly to the prior mean ($\mathbb{E}[\boldsymbol{\delta}_{c'}] = \mathbf{0}$). The effective concept smoothly and safely falls back to the universal structural prior ($\boldsymbol{\mu}_{\text{eff}, c'} = \boldsymbol{\mu}_{\text{shared}}$) without catastrophic failure.
- **Rapid Few-Shot Adaptation:** With just 1 to 3 observations of a new color c , Bayesian updating rapidly adjusts the residual $\boldsymbol{\delta}_c$, capturing color-specific physical behaviors without overwriting the global structural knowledge.

C. Inverse Planning as Gradient-Free Optimization

Given a start state $\mathbf{s}_{start}$ and an abstract goal concept C_{goal} , the planner seeks an optimal action sequence $A^* = (a_1, \dots, a_T)$. We formulate this as a greedy maximization of the log-inclusion score between the imagined future state and the dynamically instantiated effective goal concept:

1. **Context-Aware Action Generation:** Detect the dominant object color c in $\mathbf{s}_{start}$ to constrain the action space $\mathcal{A}$ (e.g., restricting the "paintbrush" to the object's native color c).
2. **Forward Simulation:** For each candidate action $a \in \mathcal{A}$, predict the hypothetical cumulative outcome using the Motor System (Section 9):

$$\Delta_{hyp} = \Delta_{accum} + \text{Imagine}(a)$$

3. **Contextual Evaluation:** We instantiate the effective goal concept adapted to the current color context c , yielding $C_{eff}^{(c)} = (\boldsymbol{\mu}_{shared} + \boldsymbol{\delta}_c, \boldsymbol{\Sigma}_{shared})$. Treating the simulation delta as a noisy observation $O_{hyp} \sim \mathcal{N}(\Delta_{hyp}, \epsilon_{obj} \mathbf{I})$, we select the action that maximizes semantic inclusion directly in the unwhitened space:

$$a^* = \operatorname{argmax}_{a \in \mathcal{A}} \left[\log_{inc} (O * hyp, C_{eff}^{(c)}) \right]$$

This mechanism allows the agent to satisfy the abstract goal "grow" using the exact visual constraints dynamically parsed from the current environment, maintaining strict probabilistic semantics in the full feature space.

11. Generative Concept Composition

(experiment_action.pdf experiment 3)

"How are new visual concepts created without observation?"

We generalize the Product of Experts mechanism (from Section 9) to arbitrary semantic composition. This provides a rigorous mathematical basis for **Generative Imagination**—the ability to hallucinate a "Blue Box" having only seen "Blue things" and "Box things."

A. The Composition Operator

Let $\mathcal{C}_{visual}$ be the manifold of all visual concepts. We define a composition operator

$$\oplus : \mathcal{C}_{visual} \times \mathcal{C}_{visual} \rightarrow \mathcal{C}_{visual}.$$

For two concepts $A(\boldsymbol{\mu}_A, \boldsymbol{\Sigma}_A)$ and $B(\boldsymbol{\mu}_B, \boldsymbol{\Sigma}_B)$, the composed concept $Z = A \oplus B$ is defined by the renormalized pointwise product of their probability density functions:

$$P(\mathbf{x} | Z) \propto P(\mathbf{x} | A) \cdot P(\mathbf{x} | B)$$

B. Parameter Synthesis

Under the diagonal Gaussian assumption, the closed-form solution for the synthesized mean and variance of Z can be computed independently for each dimension d :

Synthesized Mean:

$$\boldsymbol{\mu}_{Z,d} = \frac{\boldsymbol{\mu}_{A,d} \sigma_{A,d}^{-2} + \boldsymbol{\mu}_{B,d} \sigma_{B,d}^{-2}}{\sigma_{A,d}^{-2} + \sigma_{B,d}^{-2}}$$

Synthesized Variance:

$$\sigma_{Z,d}^2 = \frac{1}{\sigma_{A,d}^{-2} + \sigma_{B,d}^{-2}}$$

C. Emergent Construction

- **Shape Dominance:** For shape-related dimensions d (spatial occupancy), concept A ("Box") has high precision ($\sigma_{A,d}^{-2} \gg 0$) while concept B ("Blue") has low precision ($\sigma_{B,d}^{-2} \approx 0$). The formula naturally yields $\mu_{Z,d} \approx \mu_{A,d}$.
- **Color Dominance:** For color-related dimensions (Lab channels), the precision weights reverse, and the result seamlessly yields $\mu_{Z,d} \approx \mu_{B,d}$.

The resulting concept Z strictly lies within the intersection of the high-probability regions of its parents. Because the precisions add up, the resulting variance is strictly smaller than both parent variances, effectively constructing a highly specific, hallucinated prototype that satisfies both semantic constraints simultaneously.

12. Neuro-Symbolic Bayesian Program Learning (BPL)

"How does the agent learn abstract grammatical rules and generalize to unseen compositions?"

We address the **Meta-Learning Compositional (MLC)** challenge by framing concept learning not as statistical feature mapping, but as **Bayesian Program Induction**. The agent does not simply classify inputs; it infers a latent symbolic program π (Trace) that, when executed on a probabilistic stack machine, generates the observed visual outputs.

This framework bridges the gap between connectionist perception (continuous color vectors) and symbolic reasoning (discrete syntax) via a **Two-Loop Meta-Learning Architecture**.

A. The Probabilistic Generative Model (NeuroConcept)

Each word w in the lexicon is modeled as a **Probabilistic Program Module (NeuroConcept)**. It is not a static point, but a mixture of distributions parameterizing the operations of a latent Stack Machine.

Let Φ be the meta-learned global priors. The behavior of a word w is governed by a parameter tuple θ_w :

1. Syntactic Role Distribution (P_{role})

What computation does w perform? It is modeled by a Dirichlet-Multinomial distribution over the set of primitives $\mathcal{R} = \{\text{EMIT}, \text{REPEAT}, \text{SWAP_INFIX}, \text{CONCAT_INFIX}, \text{OVER_INFIX}\}$:

$$P_{cortex}(r \mid w, \text{ctx}) = \frac{\alpha_r \cdot \Psi_r(\text{ctx}) + n_{w,r}}{\sum_{r'} (\alpha_{r'} \cdot \Psi_{r'}(\text{ctx}) + n_{w,r'})}$$

- **Inductive Bias (α_r):** The global structural prior (e.g., favoring nouns, $\alpha_{\text{EMIT}} > \alpha_{\text{OVER_INFIX}}$).
- **Structural Context (Ψ_r):** A dynamic gain factor (`context_role_prior`) that modulates the prior based on the word's position and stack depth (e.g., words at the end of a sentence strongly boost the REPEAT prior).

HPC Memory Fusion(actually HPC if not helpful in this task): Under the Complementary Learning Systems (CLS) architecture, this slow cortical prior is fused with fast hippocampal retrieval signals via a log-linear model to aggressively reduce the search space:

$$\ln P_{fused}(r | w) = \ln P_{cortex}(r | w) + \lambda_{mem} \ln P_{HPC}(r | w)$$

2. Semantic Emission (P_{emit}) & Color Representations

If the role is **EMIT** (a noun), what visual value is produced? Crucially, the system supports two distinct representation modes for visual meaning, selectable via global priors:

Mode A: Discrete One-Hot Space (Default)

- **Space:** 6D one-hot vector ($d = 6$, `N_COLORS=6`).
- **Mechanism:** Modeled via a discrete Dirichlet prior over predefined color names.
- **Configuration:** `gauss=False`, `nig=NIGParams(d=6)`, and `delta` enabled (e.g., `delta={'BLUE': 1.0, ...}`).
- **Decoding:** Simply uses `argmax (vec_to_color)` to map the output vector back to a color string.

Mode B: Continuous CIELAB Space (`gauss=True`)

- **Space:** 3D continuous CIELAB perceptual space ($d = 3$).
- **Mechanism:** Modeled via a **Normal-Inverse-Gamma (NIG)** conjugate prior to handle continuous perceptual uncertainty. The concept parameter is a Gaussian distribution $\mathcal{N}(\mu, \Sigma)$. The scoring function uses the KL-divergence (or Gaussian log-likelihood) between the observed object $\mathbf{x}$ and the concept's posterior belief:

$$\text{score}(\mathbf{x}, w) = -\frac{1}{\tau_{inc}} D_{KL}(\mathcal{N}(\mathbf{x}, \epsilon_{obj} \mathbf{I}) \parallel P(\mu, \Sigma | \mathcal{D}_w))$$

The posterior parameters are derived analytically via **Weighted Moment Matching** from the soft-weighted traces (where ω_i is the responsibility weight of the i -th trace):

$$\mu_{post} = \frac{\kappa_0 \mu_0 + \sum \omega_i \mathbf{x}_i}{\kappa_0 + \sum \omega_i}, \quad \Sigma_{post} = \frac{2\beta_0 + \sum \omega_i (\mathbf{x}_i - \mu_{post})^2}{2\alpha_0 + \sum \omega_i}$$

(Note: Σ_{post} correctly centers the SSE on the posterior mean μ_{post} rather than the sample mean to rigorously account for the prior pull μ_0 .)

- **Decoding:** Cannot use `argmax`. The output 3D Lab coordinates **MUST** be mapped to the discrete color name via Nearest Neighbor mapping to pre-defined Lab prototypes (`lab_vec_to_color / nearest_color`).
- **Intuition:** The Lab mode allows the model to naturally grasp that "dark blue" and "light blue" are geometrically closer than "blue" and "red," facilitating smooth few-shot generalization under perceptual noise.

3. Latent Arity & Span (P_{span})

How many items on the stack does this operator bind? (e.g., "twice" binding to a complex phrase). This is a Latent Variable $a \in \mathbb{N}^+$. We impose a geometric "Span Prior" to favor local binding (Occam's Razor):

$$\ln P(a) = -\tau_{span} \cdot (a - 1)$$

B. The Neuro-Symbolic Execution Engine

Execution occurs on an immutable stack of vector sequences (`StackState`). A program trace π is a sequence of steps $z_t = (w_t, r_t, a_t, \theta_t)$. Unlike standard stack machines, the Arity (a_t) is inferred dynamically during parsing.

- **EMIT($\mathbf{v}$):** Pushes a sampled vector $\mathbf{v} \sim P_{emit}(\cdot | w)$.

$$S_{t+1} = S_t \cup \{[\mathbf{v}]\}$$

- **REPEAT(k, a):** Pops a items (expression X), pushes X repeated k times.

$$S_{t+1} = (S_t \setminus X) \cup \underbrace{\{X \oplus \dots \oplus X\}}_k$$

- **INFIX(a):** Pops a items (A), consumes next token (B), and pushes the composed structure (e.g., `SWAP_INFIX` $\rightarrow B, A$).

AST Decoding (Deferred Hole-Filling)

To resolve nested scope ambiguity, the inference engine constructs a latent Abstract Syntax Tree (AST). Consider the query: "3 after DAX thrice".

When `after` (`SWAP_INFIX`) opens a "hole" (argument slot), the beam search branches:

1. **Immediate Fill (Cascade):** `DAX` plugs into the hole. (Wrong scope: `thrice` then binds to the swapped result).
2. **Deferral:** Keep the hole open, add `DAX` as a new root. `thrice` wraps `DAX`, and the auto-closing mechanism eventually drops `[DAX thrice]` into the `after` hole. (Correct wide scope).

C. Abductive Inference (The E-Step)

"What program best explains the observed input-output pair?"

Given an instruction U and a target visual sequence Y , we approximate the posterior $P(\pi | U, Y)$ using **Abductive Beam Search**. The objective function for a candidate trace π is:

$$\mathcal{L}(\pi) = \ln P(\pi | U, \Phi) + \ln P(Y | \text{Exec}(\pi)) + \ln P_{S_1}(U | \text{Exec}(\pi))$$

1. Soft Visual Likelihood (Alignment)

Since the symbolic execution generates continuous vectors $\hat{Y}$, we define a differentiable **Soft Edit Distance** with a Gaussian substitution cost to handle length mismatches and visual noise. Furthermore, EMIT parsing applies a **Local Alignment Penalty** ($-\lambda |\Delta_{\text{offset}}|$) to aggressively constrain the search space:

$$\text{Cost}_{\text{sub}}(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \exp\left(-\frac{|\mathbf{y} - \hat{\mathbf{y}}|^2}{2\sigma^2}\right)$$

$$\ln P(Y | \pi) \approx -\beta \cdot \text{SoftLevenshtein}(Y, \text{Exec}(\pi)) - \lambda \sum |\Delta_{\text{offset}}|$$

2. Pragmatic Utility (RSA)

To resolve semantic ambiguity (e.g., distinguishing "blue" from "1 blue"), we inject a Speaker Utility term S_1 . We score how rationally a speaker would choose utterance U to convey the execution meaning M , competing against alternative utterances $\mathcal{A}$ (e.g., deletions or paradigmatic substitutions):

$$\mathcal{U}_{RSA}(U; M) = \alpha_{RSA} \ln P_{L_0}(M | U) - \lambda_{cost}|U|$$

$$\ln P_{S_1}(U | M) = \mathcal{U}_{RSA}(U; M) - \ln \sum_{U' \in \mathcal{A}} \exp(\mathcal{U}_{RSA}(U'; M))$$

D. Meta-Learning Algorithm (The Two Loops)

The system learns "how to learn" by optimizing priors over a distribution of episodes.

1. Inner Loop (Task Adaptation via Soft-EM)

For a specific episode with support set $\mathcal{D}_{sup}$, we refine local concept parameters θ_w .

- **E-Step:** Run beam search to find top- K traces $\{\pi_k\}$. Compute responsibilities ω_k via Softmax of scores $\mathcal{L}(\pi_k)$.
- **M-Step:** Update word parameters by accumulating weighted sufficient statistics over all steps t within the traces:

$$n_{w,r}^{(new)} \leftarrow \gamma_{decay} \cdot n_{w,r}^{(old)} + \sum_{k=1}^K \omega_k \sum_{t \in \pi_k} \mathbb{I}(\text{word}_t = w \wedge \text{role}_t = r)$$

2. Outer Loop (Prior Optimization via Empirical Bayes)

We optimize global priors Φ (e.g., $\alpha, \gamma, \tau_{span}, \beta, \lambda$) to maximize the marginal likelihood of Query examples given Support examples across multiple background episodes:

$$\Phi^* = \operatorname{argmax}_{\Phi} \sum_{ep} \ln P(Y_{query} | U_{query}, \mathcal{D}_{sup}^{(ep)}; \Phi)$$

This distills **Inductive Biases** (e.g., "words are usually nouns", "operators bind locally", "Lab variance limits") from background experience, robustly enabling **One-Shot Generalization** in completely novel environments.